

MANAGING SELINUX (BASICS)

TASKS

1. Check the current SELinux status on your Linux system using the `getenforce` and `sestatus` commands, and write down what each output means.

Check SELinux Status

Step 1: Check SELinux mode

`getenforce`

```
[root@localhost ~]# getenforce
Enforcing
```

Step 2: Get detailed status

`sestatus`

```
Enforcing
[root@localhost ~]# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:       /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:         enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     33
[root@localhost ~]#
```

Enforcing → SELinux is active and enforcing security rules.

Permissive → SELinux logs violations but does not block them.

Disabled → SELinux is turned off.

2. Switch SELinux to permissive mode temporarily using the `setenforce` command, then try creating a new file in `/var/www/html` and note any differences in error messages or permissions compared to enforcing mode.
Hint: Use `setenforce 0` for permissive and `setenforce 1` to return to enforcing.

Switch to Permissive Mode

Step 1: Become root

su -

Step 2: Change to permissive mode

setenforce 0

Verify:

getenforce

```
[root@localhost ~]# su -  
[root@localhost ~]# setenforce 0  
[root@localhost ~]# getenforce  
Permissive
```

Step 3: Create a file

touch /var/www/html/testfile.txt

Verify:

ls -l /var/www/html

```
[root@localhost ~]# touch /var/www/html/testfile.txt  
[root@localhost ~]# ls -l /var/www/html  
total 0  
drwxr-xr-x. 2 root  root  43 Jul 14 12:04 myflipkart  
drwxr-xr-x. 2 apache apache 24 Jul 14 13:45 mysite  
-rw-r--r--. 1 root  root   0 Jul 14 14:19 testfile.txt
```

Step 4: Return to enforcing mode

setenforce 1

Verify:

getenforce

```
[root@localhost ~]# setenforce 1  
[root@localhost ~]# getenforce  
Enforcing
```

3. List all files in /var/www/html and use the ls -Z command to display their SELinux security contexts. Change the context of one file to httpd_sys_content_t using chcon and verify the

change.

Hint: Use `chcon --type=httpd_sys_content_t filename</em>`

View and Change SELinux Context

Step 1: List SELinux contexts

`ls -Z /var/www/html`

```
[root@localhost ~]# ls -Z /var/www/html
unconfined_u:object_r:httpd_sys_content_t:s0 myflipkart
unconfined_u:object_r:httpd_sys_content_t:s0 mysite
unconfined_u:object_r:httpd_sys_content_t:s0 testfile.txt
[root@localhost ~]#
```

Step 2: Change context

`chcon --type=httpd_sys_content_t /var/www/html/testfile.txt`

Step 3: Verify

`ls -Z /var/www/html/testfile.txt`

```
[root@localhost ~]# ls -Z /var/www/html/testfile.txt
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/testfile.txt
[root@localhost ~]#
```

4.A

**user is unable to upload images to a folder for their food review blog (like Zomato) hosted in /srv/foodie/images. Use the audit2why and audit2allow tools to analyze SELinux denial logs and suggest a solution to allow uploads.

Hint: Check /var/log/audit/audit.log for recent denials.**

SELinux Troubleshooting Scenario

Scenario

A user cannot upload images to:

/var/www/html/foodie/images

Step 1: Create folder

`mkdir -p /var/www/html/foodie/images`

Step 2: Check SELinux context

ls -Zd /var/www/html/foodie/images

```
[root@localhost ~]# mkdir -p /var/www/html/foodie/images
[root@localhost ~]# ls -Zd /var/www/html/foodie/images
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/foodie/images
```

Step 3: Check audit log

tail /var/log/audit/audit.log

```
bash: step: command not found...
[root@localhost ~]# tail /var/log/audit/audit.log
type=SERVICE_START msg=audit(1784019005.415:878): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=SERVICE_STOP msg=audit(1784019022.737:879): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=SERVICE_START msg=audit(1784019061.689:880): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=SERVICE_STOP msg=audit(1784019079.201:881): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=MAC_STATUS msg=audit(1784019089.812:882): enforcing=1 old_enforcing=0 auid=0 ses=9 enabled=1 old-enabled=1 lsm=selinux res=1AUID="root"
type=SYSCALL msg=audit(1784019089.812:882): arch=c000003e syscall=1 success=yes exit=1 a0=3 a1=7ffc27674a30 a2=1 a3=1 items=0 ppid=16018 pid=16121 auid=0 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=9 comm="setenforce" exe="/usr/sbin/setenforce" subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(null)ARCH=x86_64 SYSCALL=write AUID="root" UID="root" GID="root" EUID="root" SUID="root" FSUID="root" EGID="root" SGID="root" FSGID="root"
type=PROCTITLE msg=audit(1784019089.812:882): proctitle=7365746556E666F7263650031
type=SERVICE_START msg=audit(1784019583.680:883): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=SERVICE_START msg=audit(1784019597.573:884): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=packagekit comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=SERVICE_STOP msg=audit(1784019601.210:885): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=NetworkManager-dispatcher comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
```

OR

grep denied /var/log/audit/audit.log

```
[root@localhost ~]# grep denied /var/log/audit/audit.log
[root@localhost ~]#
```

Step 4:

Fix SELinux context

chcon -R -t httpd_sys_rw_content_t /var/www/html/foodie/images

Verify:

`ls -Zd /var/www/html/foodie/images`

```
[root@localhost ~]# chcon -R -t httpd_sys_rw_content_t /var/www/html/foodie/images
[root@localhost ~]# ls -Zd /var/www/html/foodie/images
unconfined_u:object_r:httpd_sys_rw_content_t:s0 /var/www/html/foodie/images
```

Problem:

The web server could not upload images because SELinux did not allow write access.

Solution:

Changed the SELinux context to:

`httpd_sys_rw_content_t`

This allows Apache (`httpd`) to write files into the directory.

